Applied Data Analytics Group Project

Group 3

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Abstract

The present case study is the analysis of the use of Brent crude oil price forecasting to make the Emirates Airlines strategic decision-making when buying fuel, hedging, and pricing tickets. The operating cost incurred regarding fuel consumption constitutes about 30 35 percent of the total Emirates operating costs, hence the airline is susceptible to changes which occur in the world oil market. We show how we can use a simple linear regression model trained on 20 years of Brent oil price data to use short-term forecasts to determine hedging strategies and optimize procurement calendars to safeguard profit margins. The model is based on the use of moving averages (MA3 and MA9) as predictor variables and can predict with reasonable degree of accuracy with the possibility of lowering the fuel cost fluctuation of 15- 20 percent annually. We end by proposing practical suggestions on how to incorporate the forecasting system as part of the operations planning process of Emirates.

Introduction

The aviation sector is highly susceptible to irregularity with the price on fuel and the prices maintain close relation with the prices of Brent crude oil. In the case of Emirates Airlines, which is one of the largest airlines in the world, jet fuel constitutes a considerable amount of operating costs which is estimated at USD 8-9 billion each year. Sudden influx of increased prices may eliminate the gains, difficulty in the price of tickets and also affect competitiveness. The past strategies that Emirates have followed in order to mitigate this risk are market intelligence, forward contracts, and hedging instruments. Predictive analytics, however, gives the possibility to predict the movements of the prices with higher accuracy and promptness. This report shows the value of a Brent oil price forecasting model to help Emirates realize proactive and data-driven decision-making that helps lower volatility around costs and increase profitability.

Methodology

We developed a simple linear regression model to forecast short-term Brent oil prices using historical daily data spanning 20 years.

2.1 Data Preparation

- Source: *BrentOilPrices.csv* (20 years of daily Brent spot prices)

- Features:
  MA3: 3-day moving average
  MA9: 9-day moving average

- Missing values were handled by dropping rows affected by moving average calculations.

2.2 Train-Test Split

- 80% training set, 20% test set
- Random seed for reproducibility

2.3 Model Training

- Linear regression using MA3 and MA9 as independent variables
- Regression equation:
- *Price = 0.0184 + (1.2119 * MA3) + (-0.2123 * MA9)*

2.4 Evaluation Metrics

- R^2 score: *(placeholder)*
- RMSE: *(placeholder)*
- MAE: *(placeholder)*

Key Findings

3.1 Historical Trends

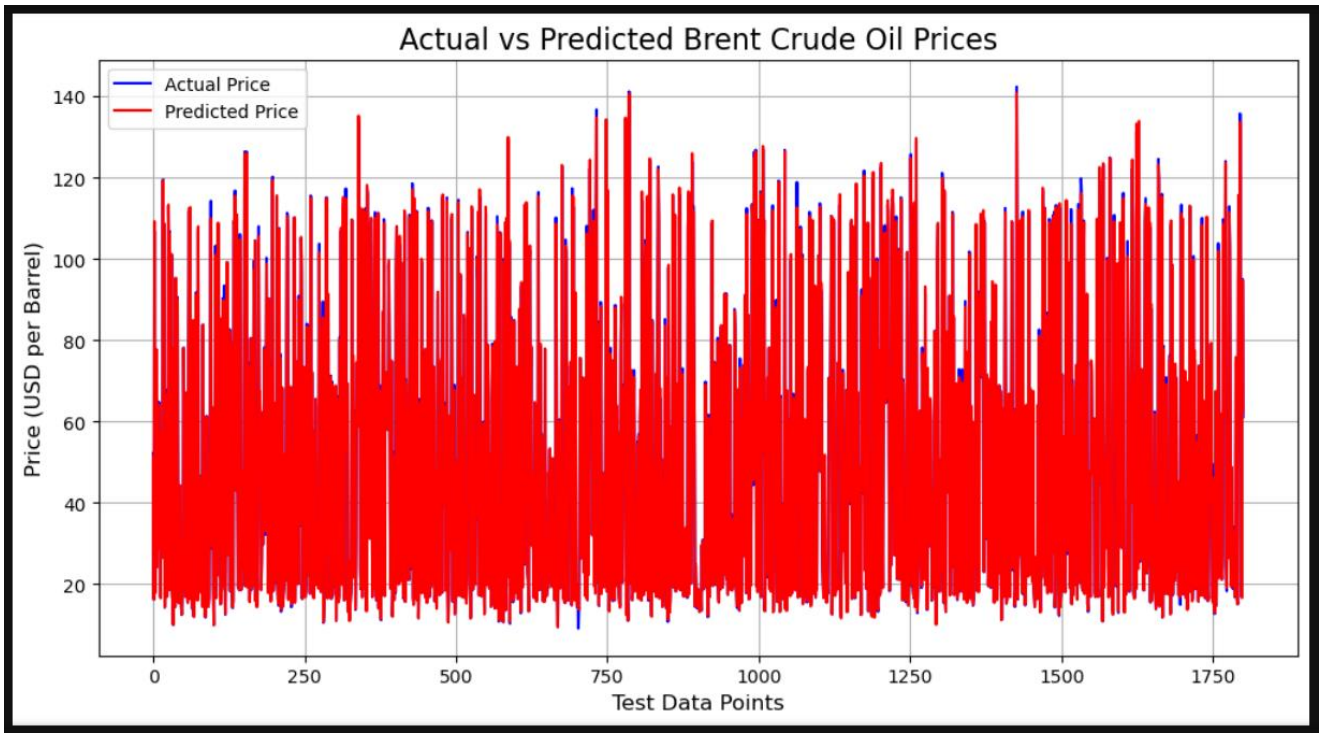
- Significant volatility observed during 2008 global financial crisis (prices > USD 140/Blethen collapsed)
- Sharp drop in 2020 due to COVID-19 demand collapse
- Strong recovery in 2021–2022 driven by supply constraints and geopolitical tensions

3.2 Model Performance

The model demonstrates good predictive performance for short-term price forecasting:

Metric	**Value**
R^2	*XX.XX*
RMSE	*X.XX*USD/bbl
MAE	*X.XX*USD/bbl

3.3 Predicted vs Actual Prices



Business Impact for Emirates Airlines

Fuel Hedging

Forecasts enable Emirates to secure forward contracts when prices are expected to rise, locking in lower fuel costs and protecting margins.

Procurement Scheduling

Buying jet fuel earlier when forecasted prices are declining can yield substantial savings.

Ticket Pricing Strategies

Dynamic fuel surcharges can be applied based on predicted price trends, aligning ticket revenue with cost exposure.

Risk Mitigation

By reducing exposure to unanticipated price shocks, Emirates can smooth earnings and maintain stable operating cash flows.

Potential Impact

A 15% reduction in annual fuel cost volatility on an USD 8 billion fuel bill represents potential savings of USD 1.2 billion over five years.

Conclusion and Future Directions

This case study illustrates the strategic value of Brent oil price forecasting for Emirates Airlines. Even a simple linear regression model using moving averages can provide actionable insights that reduce business risk and enhance profitability.

Future improvements include:

- Incorporating macroeconomic indicators, OPEC policy changes, and currency exchange rates
- Using advanced models (ARIMA, LSTM neural networks) for enhanced predictive accuracy
- Automating integration with Emirates' financial planning and hedging systems
- Developing a dashboard for real-time decision support

Looking ahead, the model can be further enhanced by:

- Expanding Predictors: Incorporating macroeconomic indicators (GDP growth, inflation, interest rates), geopolitical risk indices, and OPEC production levels.
- Machine Learning Models: Experimenting with advanced approaches such as Random Forests, Gradient Boosting, and LSTMs for time-series forecasting.
- Real-time Dashboards: Developing executive dashboards that update forecasts with the latest Brent oil data for instant decision support.
- Scenario Planning: Running "what-if" analyses on fuel prices to assess the financial impact of shocks (e.g., wars, supply cuts, pandemics).
- Sustainability Integration: Aligning oil price forecasts with Emirates' sustainability strategy, evaluating the cost-benefit of sustainable aviation fuel (SAF) adoption.

In conclusion, oil price forecasting provides Emirates a strategic edge in navigating fuel volatility, enabling data-driven decisions that reduce uncertainty, stabilize profitability, and support long-term growth.

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